

Exercice 1

1/ $f \in [20 \text{ Hz} ; 20 \text{ kHz}]$

2/ Ces fréquences correspondent à des longueurs d'ondes

$$\lambda = \frac{c}{f} = \left| \begin{array}{l} \frac{3 \cdot 10^8}{20} = 15\,000 \text{ km} \\ \frac{3 \cdot 10^8}{20 \cdot 10^3} = 15 \text{ km} \end{array} \right.$$

Il faudrait une antenne de 15 000 km!

3/ $v_s = v_p(t) + k v_e(t) v_p(t)$

4/ $v_s = A_p \cos(2\pi f_p t) + k A_p \cos(2\pi f_p t) \cdot A_e \cos(2\pi f_e t)$

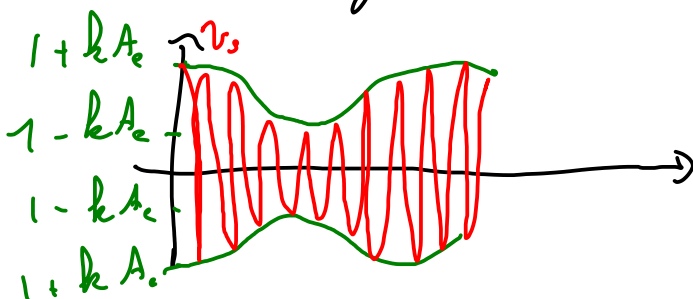
$$= A_p \cos(2\pi f_p t) [1 + k A_e \cos(2\pi f_e t)]$$

2 cas :

$$k A_e < 1$$

$$1 + k A_e \cos(2\pi f_e t)$$

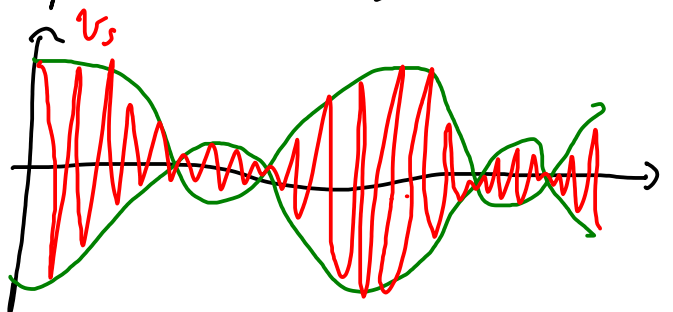
est toujours > 0



$$k A_e > 1$$

$$1 + k A_e \cos(2\pi f_e t)$$

peut être 0

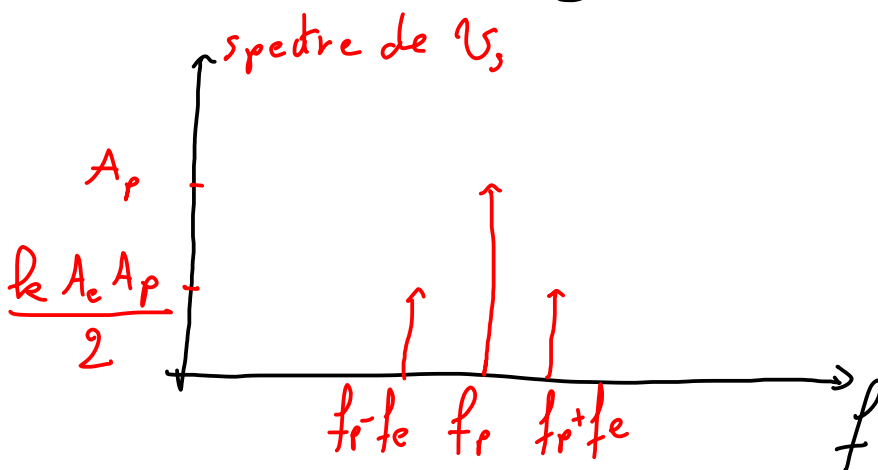


On peut démoduler

On ne peut pas démoduler

→ il faut $k < \frac{1}{A_e}$

$$5 / v_s(t) = A_p \cos(2\pi f_p t) + \frac{k A_e A_p}{2} \left[\cos(2\pi(f_p + f_c)t) + \cos(2\pi(f_p - f_c)t) \right]$$



6/ Si v_e a pour spectre



alors v_s a pour spectre

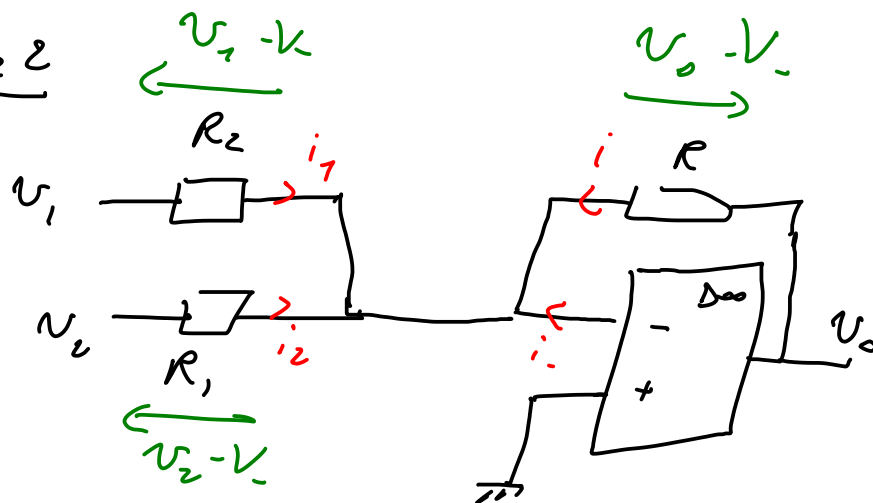


7/ Le spectre de V_s occupe 40 kHz sur le spectre.

Dans la bande $[520; 1620]$ kHz, on peut en faire tenir $\frac{1620 - 520}{40} \approx 27$ canaux

Exercice 2

1/



Kirchhoff's law : $i_1 + i_2 + i + i_- = 0$

but $i_- \approx 0$

$$\rightarrow i_1 + i_2 + i = 0$$

$$\rightarrow \frac{1}{R_1} (v_1 - V_-) + \frac{1}{R_2} (v_2 - V_-) + \frac{1}{R} (v_0 - V_-) = 0$$

$$\rightarrow V_- \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R} \right) = \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_0}{R}$$

$$\rightarrow V_- = \frac{\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_0}{R}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R}}$$

2/ $\begin{cases} \text{ILA ideal} \\ \text{negative retroaction} \end{cases}$

$$\rightarrow \varepsilon = 0$$

$$\rightarrow V_- = V_+ = 0$$

$$\rightarrow \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_o}{R} = 0$$

$$\rightarrow v_o = -R \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} \right)$$

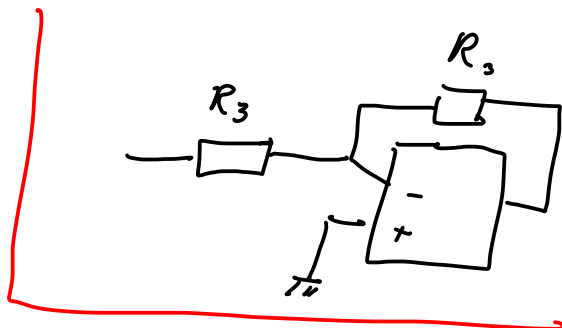
3/ If $R = R_1 = R_2$

$$v_o = -v_1 - v_2$$

4/ $H_2 = \frac{v_{s2}}{v_2} = \frac{v_1 + v_2}{-v_1 - v_2} = -1$

An inverting amplifier can have this transfer function:

$$H_2 = -\frac{R_3}{R_4} \rightarrow \text{if } R_3 = R_4 \rightarrow H_2 = -1$$

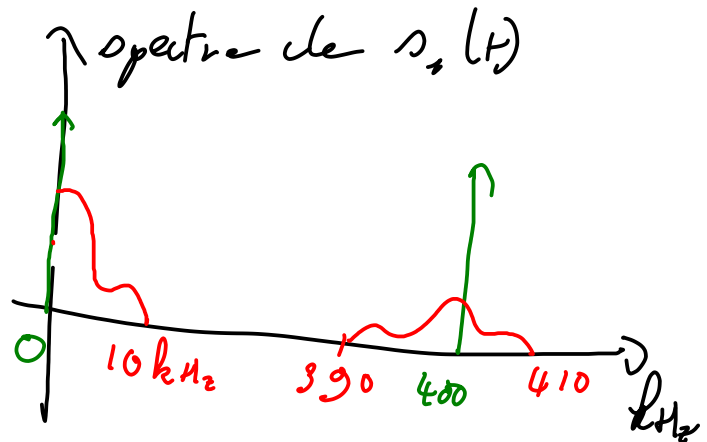
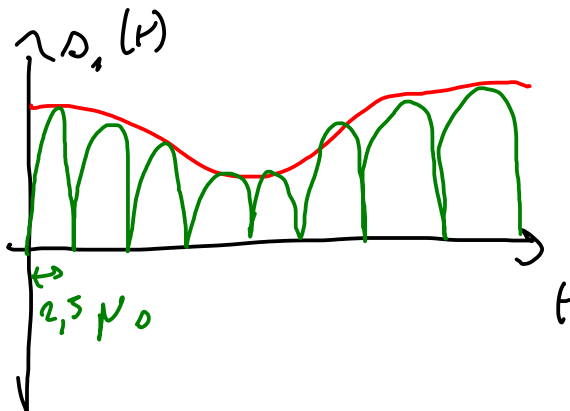
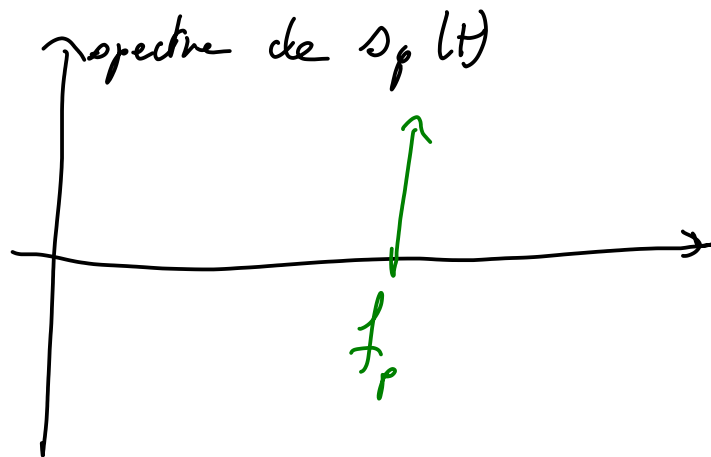
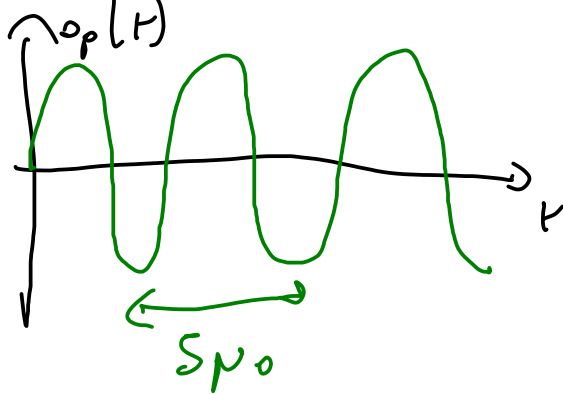
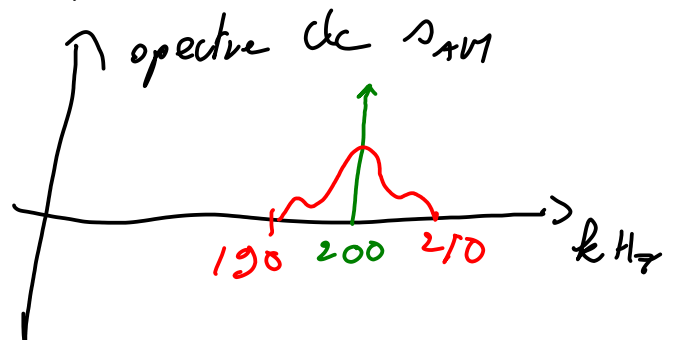
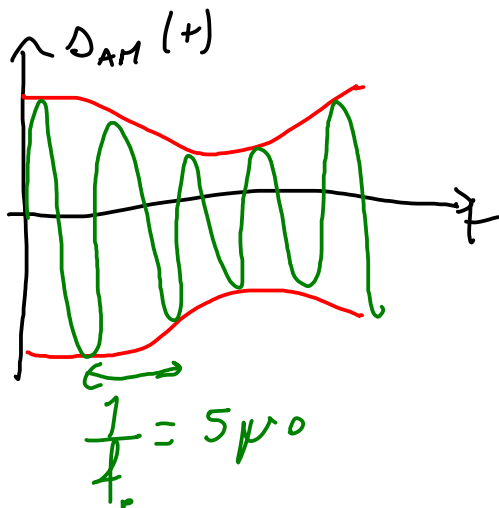


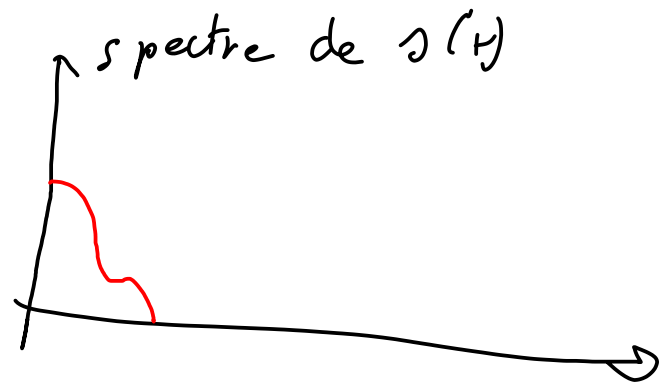
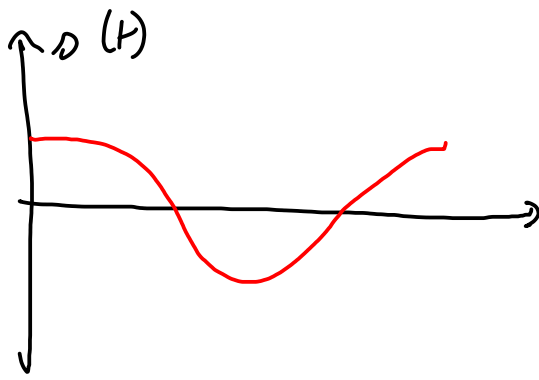
Exercice 3

$$1/ \quad s_{AM}(t) = A_p \cos(2\pi f_p t) (1 + k_D(t))$$

$$D_p(t) = A_p \cos(2\pi f_p t)$$

$$D_s(t) = A_p^2 \cos^2(2\pi f_p t) (1 + k_D(t))$$



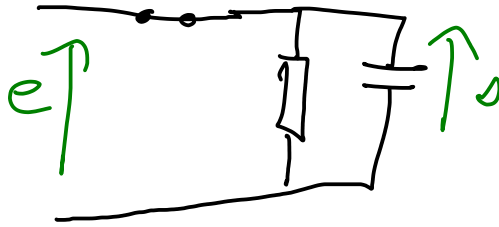


2/ On peut prendre $f_c = 16 \text{ kHz} = \frac{1}{2\pi RC}$

avec par exemple $R = 1 \text{ k}\Omega$
 $C = 10 \text{ nF}$

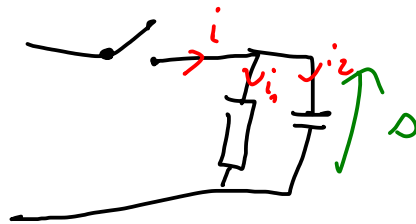
Exercice 4

1/



$$\Delta(t) = e(t)$$

2/



$$0 = i_1 - i_2 \quad \text{or} \quad i_1 = i_2$$

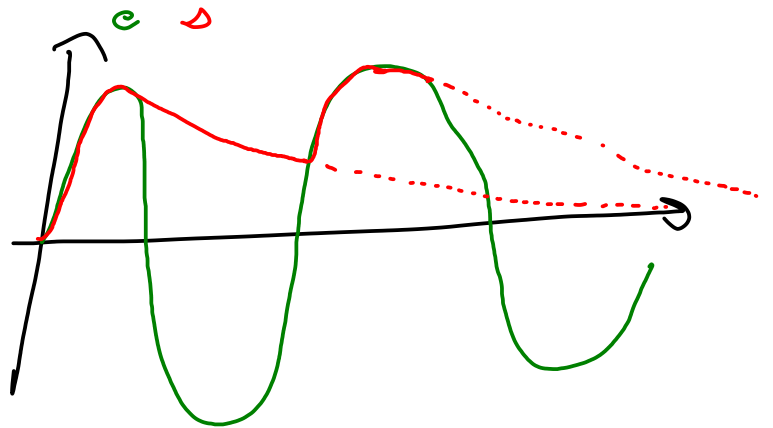
$$\rightarrow i_1 = -i_2$$

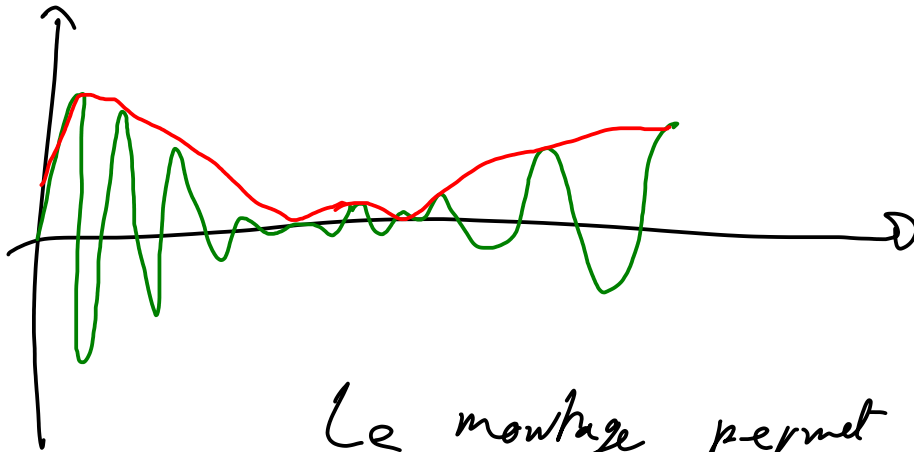
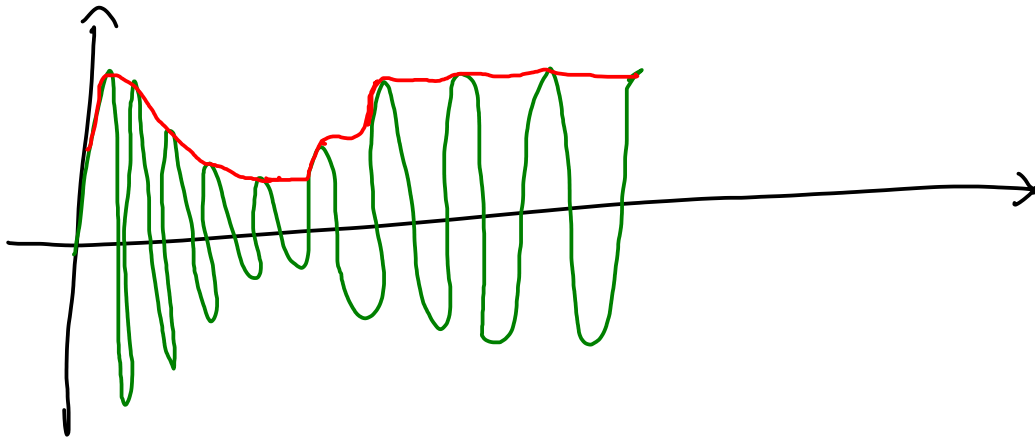
$$\frac{\Delta}{R} = -C \frac{d\Delta}{dt}$$

$$\rightarrow \frac{d\Delta}{dt} + \frac{1}{RC} \Delta = 0$$

$$\rightarrow \Delta(t) = A e^{-\frac{t}{RC}}$$

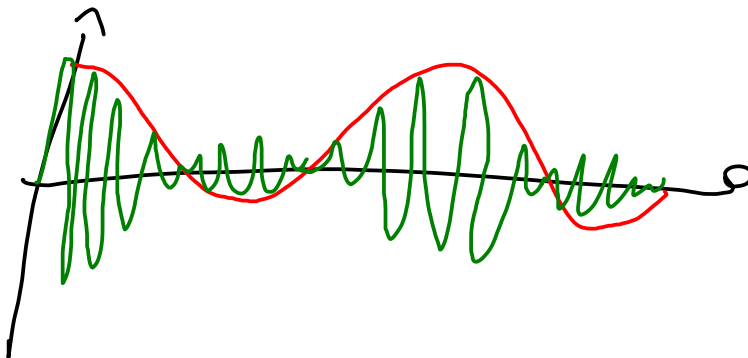
3/ En zoom :





Le montage permet de détecter
(l'enveloppe)

4/ Pour le signal 2, on aurait pensé à
un signal modulant sinusoïdal :



on n'obtient pas ce signal avec
le détecteur d'enveloppe.

Exercice 5

$$1/ \quad \frac{\Delta f}{f} = \frac{2v}{c}$$

$$\Rightarrow \Delta f = 2 \cdot \frac{90 \times 1000}{3600} \cdot \frac{1}{3 \cdot 10^8} \cdot 24 \cdot 10^9$$
$$= \underline{4 \text{ kHz}}$$

Difficile de mesurer $24\,000\,000\,000 \text{ Hz}$
précisément pour faire la différence avec
 $24\,000\,000\,000 \text{ Hz}$.

$$2/ \quad D_r(t) = A_r \cos(2\pi f_r t)$$

$$\text{avec } f_r = 24 \text{ GHz} + 4 \text{ kHz}$$

$$= f + \Delta f = \underline{f \left(1 + \frac{2v}{c}\right)}$$

$$3/ \quad D \cdot D_r = A \cdot A_r \cos(2\pi f t) \cos(2\pi f_r t)$$

$$= \frac{A A_r}{2} \left[\cos(2\pi (f_r + f) t) \right. \\ \left. + \cos(2\pi (f_r - f) t) \right]$$

$$= \frac{A A_r}{2} \left[\cos\left(2\pi f \left(2 + \frac{2v}{c}\right) t\right) \right. \\ \left. + \cos\left(2\pi f \cdot \frac{2v}{c} t\right) \right]$$

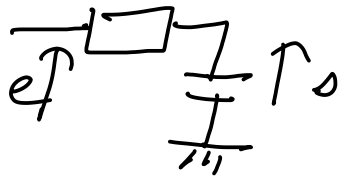
après le filtre passe-bas :

$$\frac{AA_r}{2} \cos\left(2\pi f \cdot \frac{2v}{c} t\right)$$

on peut en mesurer la fréquence

$f \cdot \frac{2v}{c}$ (ici 4 kHz) puis en déduire v .

4 /



par exemple :

$$R = 1 \text{ k}\Omega$$

$$C = 10 \text{ nF}$$

$$\rightarrow f_{\text{coupe}} = 16 \text{ kHz}$$

$$4 \text{ kHz} < f_{\text{coupe}} < 48 \text{ GHz}$$

on coupe $\cos\left(2\pi f\left(2 + \frac{2v}{c}\right)t\right)$

et on garde $\cos\left(2\pi f \frac{2v}{c} t\right)$.